

New Words and Expressions

1. Bell Laboratory 贝尔实验室，总部位于美国新泽西州的默里·希尔，是晶体管、激光器、太阳能电池、发光二极管、数字交换机、通信卫星、电子数字计算机、蜂窝移动通信设备、长途电视传送、仿真语言、有声电影、立体声录音以及通信网等许多重大发明的诞生地；自 1925 年以来，贝尔实验室共获得两万五千多项专利，现在平均每个工作日获得三项多专利；贝尔实验室同时还培养了多名诺贝尔奖获得者。

2. space vehicle power supplies 空间飞行器（如卫星、火箭、空间站等）电源。

3. terrestrial application 地面应用。

4. ubiquitous *a.* 普遍存在的；无所不在的。

5. inexhaustible *a.* 用不完的；取之不尽、用之不竭的。

6. transistor *n.* 晶体管，是一种固体半导体器件，可以用于检波、整流、放大、开关、稳压、信号调制和许多其他功能。

7. integrated circuit 集成电路，是一种微型电子器件或部件，采用一定的工艺把一个电路中所需的晶体管、二极管、电阻、电容和电感等元件及布线互连一起，制作在一小块或几小块半导体晶片或介质基片上，然后封装在一个管壳内，成为具有所需电路功能的微型结构。

8. Czochralski *n.* 丘克拉斯基法，一种晶体提拉法，是一种直接熔化原料，然后利用种晶从熔体中提拉出单晶体的方法。

9. block casting 块模浇铸法，即将熔体直接浇铸进块体模具中成型的方法。

10. polycrystalline silicon 多晶硅，单质硅的一种形态，其高温熔融状态下，具有较大的化学活泼性，能与几乎任何材料作用，具有半导体性质，是极为重要的优良半导体材料；多晶硅又是生产单晶硅的直接原料，是当代人工智能、自动控制、信息处理、光电转换等半导体器件的电子信息基础材料。

11. band structure 能带结构。

12. valence band 价带，指半导体或绝缘体在 0K 时能被电子占满的最高能带。对半导体而言，此能带中的能级基本上是连续的，全充满的能带中的电子不能在固体中自由运动；但若该电子受到光照，它可吸收足够能量而跳入下一个允许的最高能区，从而使价带变成部分充填，此时价带中留下的电子可在固体中自由运动。

13. conduction band 导带，指半导体或是绝缘体材料中一个电子所具有能量的范围，这个能量的范围高于价带 (valence band)，而所有在传导带中的电子均可经由外在的电场加速而形成电流。

14. amorphous Silicon 非晶硅，单质硅的一种形态，不具有完整的金刚石型晶胞，纯度不高，熔点、密度和硬度等数值也明显低于晶态硅；化学性质比晶态硅活泼。非晶硅可由活泼金属（如钠、钾等）在加热下还原四卤化硅（ SiF_4 或 SiCl_4 ）或在高温下用碳或镁等还原剂与二氧化硅作用制得。

15. light-induced degradation 光致衰减效应，指太阳能电池板的效率会随着光照时间增加而降低的现象。

16. facade application 幕墙应用。

17. compound semiconductors 化合物半导体.
18. Copper Indium Diselenide 铜铟硒化合物, 该材料制作的太阳电池具有生产成本低、污染小、不衰退、弱光性能好等显著特点, 光电转换效率居各种薄膜太阳电池之首, 接近于晶体硅太阳电池, 而成本只是它的三分之一, 被称为下一代非常有前途的新型薄膜太阳电池.
19. cadmium Telluride 碲化镉, 缩写 CdTe, 位于 II-VI 位的半导体, 其禁带空隙值为 1.45eV, 有很好的光吸收性, 应用于超薄太阳能电池板.
20. ternary compound 三元化合物.
21. degradation *n.* 衰减, 退化, 降级.
22. ribbon silicon 带硅.
23. Vanguard satellite 先锋卫星.
24. breakthrough *n.* 突破; 突破性进展.
25. oilshock *n.* 石油危机; 石油冲击.
26. alternative *n.* 替代品; 供替代的选择.
27. grid-connected system 并网发电系统, 不经过蓄电池储能, 通过并网逆变器直接将电能输入公共电网的发电系统.
28. roof program 屋顶计划.

Notes

① Solar cell technology benefited greatly from the high standard of silicon technology developed originally for transistors and later for integrated circuits.

参考译文: 太阳电池技术从最初发展的晶体管及后来发展的集成电路高标准硅技术中受益匪浅。

② Theoretically, this can be explained by the semiconductor band structure of silicon in which the valence band maximum is offset from the conduction band minimum.

参考译文: 从理论上讲, 这可以从硅的半导体能带结构得到解释, 即价带最大值与导带最小值之间的偏差。

③ In spite of the complicated manufacture and the high cost, crystalline silicon still dominates the market today and probably will continue to do so in the immediate future.

参考译文: 晶体硅尽管制造复杂并且成本高昂, 但在当今市场仍然起主导作用, 而且在不久的将来可能会持续下去。

④ The breakthrough for terrestrial photovoltaics can be traced directly to the oilshock of 1973/74.

参考译文: 地面光伏应用取得突破可以直接追溯到 1973~1974 年的石油危机。

1.3 The Future of Photovoltaics

1. Boundary Conditions for the Future Development of Photovoltaics

The cost development of conventional electricity is a very important boundary condi-

tion for the future use of photovoltaics. Generally, it can be assumed that the cost of grid electricity will slowly rise over the next decades. The slope of this rise depends on the following influences: degree of liberalization of electricity markets, margins for the security of supply, proportion of distributed generation, e. g. , **cogeneration**, environmental restrictions, and last but not least, the fate of atomic energy. Practically all these influences, with the exception of liberalization, point toward higher cost of electricity. Another influence comes from the development of storage technologies. Presently, storage of electricity is not a technical, but an economic problem. Large-scale, grid-connected storage is only possible with pumped hydro installations, which are very limited in their potential. **Storage batteries** are widely used in stand-alone and mobile systems, but are much too expensive for storing grid electricity. If the full potential of photovoltaics is to be realized, better means of storage are necessary. The hydrogen economy would be a solution to the storage problem, as will be pointed out later.

The liberalization of the electricity market that is going on worldwide will influence the penetration of grid-connected PV. The following tendencies can be recognized:

—Liberalization will lead to more economics, i. e. , lower electricity prices, but this may be limited to large-scale consumers. Therefore, the overall consequence could be neutral relative to the expansion of PV in distributed systems.

—Electricity prices will reflect the real cost of generation. Prices will fluctuate widely depending on the time of day and season. Since a large part of solar electricity is generated during the high price period, this will be beneficial for PV.

—The cost of reserve capacity will lead to a lower security of supply. The role of PV as a back-up source will be favorable for this market.

2. Cost and Market Development of Stand-Alone and Grid-Connected Systems

The future of PV depends mainly on the cost development for modules and entire systems^①. For modules, which are the most expensive part of a system, has been well established for more than twenty years. It predicts a 20% price reduction for every doubling of cumulated production. Assumptions have to be made about market growth for such a prediction. If present support mechanisms are maintained, the current growth rate will continue into the future. An estimate for the market development is given by W. Hoffmann: 30% per year until 2010 and 25% thereafter.

Future development of markets: The grid-connected market will remain the most important sector, but the other sectors, remote industrial, developing countries and consumer products together reach almost the same size. These markets are, of course, interdependent. Only by price reductions due to expanding grid-connected markets can the other sectors grow accordingly.

3. PV in a Future Liberalized and Partly Decentralized Energy System

Future electricity grids will have a mixture of central and decentralized generation capacity. Central conventional and renewable plants will cooperate with small and medium

distributed generation. Modern electronics will permit a high degree of control, leading to a rapid adjustment of the economic optimum. Very important in this concept is the possibility of local storage of electricity.

Present models of a distributed energy system envision PV at many rooftops or similar structures interacting with an electricity grid and other local generators and consumers. Demand can be adjusted to some degree by central control. Besides PV, other small generators may be connected to the grid like combined heat and power plants, some based on biofuels, fuel cells, and local storage in batteries or more likely by hydrogen. Central plants can be planned by weather forecasts predicting wind and sunshine. Different **renewable energies** acting together have much lower fluctuation than each one by itself. A considerable proportion of local generation and storage may reduce the cost of distribution grids and enhance the security of supply. Much work remains to be done in working out grid and control structures for such a scenario. Undoubtedly, PV will play a major role in such a new electricity scene. The Edison project supported by the German Ministry of Economics and Technology has the goal of developing and demonstrating modular intelligent systems for such distributed grids. It involves, in particular, the combination of electricity grid and communication systems. The structure of such a future electricity distribution network is shown in Fig. 1. 2.

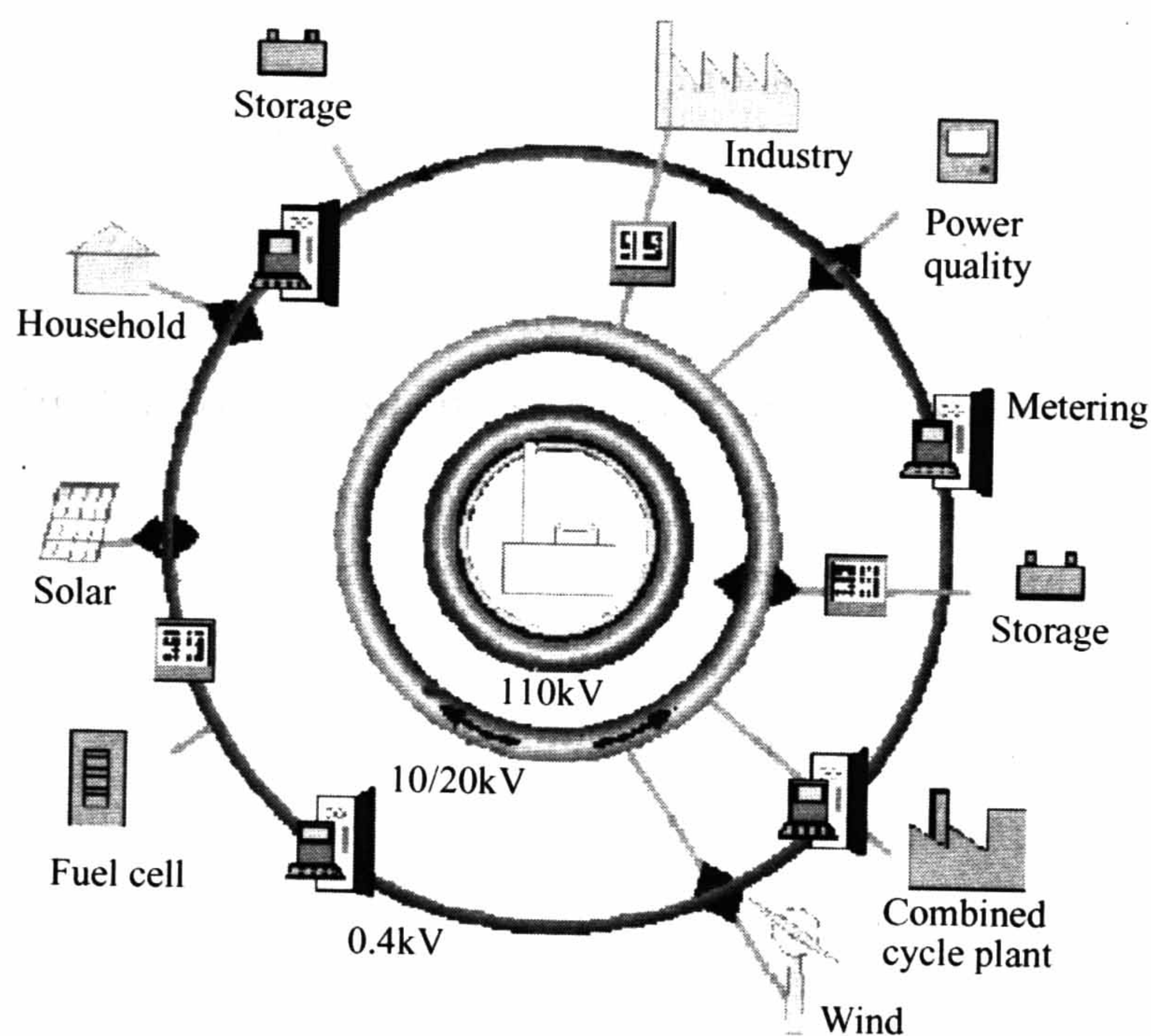


Fig. 1. 2 Structure of a future electricity distribution network

A fully autonomous system has already been demonstrated as the **Self-Sufficient Solar House**. In this project, most of the above mentioned techniques were already incorporated and tested. When they will become reality is mainly a question of cost. Fuel cells are under very intense development for mobile applications which are more demanding than stationary ones. If fuel cells become more available, local storage of hydrogen will make distributed systems more independent. From today's point of view, complete autonomy is not a

very likely option. But the grid is already in place, so why not use its advantages. If a dwelling with local generation is grid-connected, it can deliver surpluses into the grid and local storage does not have to cover extreme situations. Energy stored locally can be fed into the grid at times of peak power and fetch a higher price. Such a dwelling would have a low heating demand that would be covered by a combined heat and power plant, which could, besides covering local demand, feed energy into the grid when demand is high.

A further scenario developed some time ago recognizes the fact that as energy efficiency in buildings improves the center of demand shifts to mobility. The engine of an automobile is, after all, a combined heat and power plant.

4. PV in a Centralized Energy System

PV could in a very long-term scenario also play a role in a centralized energy system. Large PV power plants in the multi-megawatt range can be envisioned to produce electricity in forever sunny desert areas or even in outer space. Such scenarios require very low but still possible cost for modules and systems. It is quite clear that these visions can only become reality in the distant future, maybe around the middle of this century.

Deserts are regions of abundant sunshine but of very little other use. It is tempting to imagine large-scale solar energy installations in such areas where they are much more efficient than in the areas where most energy is consumed. The big challenge is to transport the energy over large distances to the consumers. Two main techniques can be identified today: high tension power lines and conversion to hydrogen.

Transmission of electricity by power lines over long distances is technically possible today. High voltage dc transmission works with low losses and has been operational for many years. Such lines could be set up, for instance, between Northern Africa and Europe or from the Gobi desert to Japan. An even further reaching idea is to establish a worldwide electricity grid. Then the storage problem could be solved very elegantly, because the sun is always shining in one half of the world. And seasonal differences can also be overcome by energy exchange between the northern and southern hemispheres. Such grids could also be used for wind energy, since many areas with very good wind resources are located in remote parts of the world. The cost of transmission has recently been estimated for transport from north Africa to central Europe. A 5000km transmission line has 18% loss at 600 kV and 14% loss at 800 kV. The cost of the losses is estimated at 0.5 cts per kW · h and the cost of transmission at 0.5 to 1.0 cts per kW · h.

The second option for energy transport is hydrogen. Water is split into hydrogen and oxygen by electrolysis and hydrogen can then serve as the energy carrier^②. Transport of hydrogen can be accomplished in three different ways:

- Hydrogen is liquefied near a harbor and then transported away by tankers.
- Hydrogen is pressurized and piped to the consumers by pipeline.
- The third pathway is the least explored and its reliability is uncertain: Hydrogen is chemically reacted to form a liquid energy carrier that can be transported, as outlined

above, for liquid hydrogen, but does not require refrigeration. At the end of the transport chain, the compound is decomposed and the hydrogen used for energetic purposes. The **hydrating reactant** can be transported back to the site of electrolysis and reused. Several chemical reactions for the described process have been identified, but no large-scale demonstration has been undertaken so far.

If a hydrogen economy should develop in the future, PV could fit very well into this scenario. A problem is the energy input for electrolysis, refrigeration, and transport of the hydrogen, which makes this pathway less efficient than direct transport of electricity, as described above. However, hydrogen can serve as a storage medium for energy and is also suitable for mobile applications.

The **Solar Power Satellite** concept was first introduced by Dr. Peter Glaser of Arthur D. Little, Inc., in 1968. According to this proposal, special satellites with very large areas of photovoltaic modules would be placed in a **geostationary orbit**. Each satellite would have an area of around 50km^2 . The electricity generated would be transmitted to the surface of the earth by antenna systems using microwave or laser beams. This concept has remained a utopia, although it is taken up periodically. Indeed, it does have attractive features: Sunlight is available in this orbit almost continually and uninterrupted by clouds or atmosphere^③. Estimates are that a solar cell in space would generate almost eight times more electricity than on earth. Just as large are the obstacles: The cost of lifting large numbers of modules into space appears prohibitive. Also, beaming powerful microwave beams to the earth causes environmental concerns. A newer plan proposes manufacturing solar cells on the moon or on an asteroid in order to avoid the gravity of the earth. Obviously, a solar power satellite is technically possible, but not realizable in the foreseeable future.



New Words and Expressions

1. cogeneration *n.* 热电发电.
2. storage battery 蓄电池.
3. stand-alone system 独立发电系统, 也称离网发电系统, 指与公共电网没有连接的光伏发电系统.
4. decentralized energy system 分散能量系统.
5. renewable energy 可再生能源, 指在自然界中可以不断再生、永续利用的能源, 具有取之不尽、用之不竭的特点, 主要包括太阳能、风能、水能、生物质能、地热能和海洋能等.
6. self-sufficient solar house 自给自足的太阳房.
7. hydrating reactant 补水反应物.
8. solar power satellite 太阳能发电卫星, 即利用空间太阳能发电, 最早是由美国工程师 Glaser 提出的, 其基本原理是空间电站利用铺设在巨大平板上的亿万片太阳电池,

在太阳光照射下产生电流，将电流集中起来，转换成无线电微波，发送给地面接收站，地面接收后，将微波恢复为直流电或交流电，就可送给用户使用。

9. geostationary orbit 地球静止轨道。

Notes

① The future of PV depends mainly on the cost development for modules and entire systems.

参考译文：光伏的未来主要取决于组件和整个系统的成本发展。

② Water is split into hydrogen and oxygen by electrolysis and hydrogen can then serve as the energy carrier.

参考译文：通过电解将水分离为氢气和氧气，然后氢气可以作为能量的载体。

③ Indeed, it does have attractive features: Sunlight is available in this orbit almost continually and uninterrupted by clouds or atmosphere.

参考译文：事实上，太阳能发电卫星有许多吸引人的特征，如在地球静止轨道获得的太阳光几乎是连续的、不间断，因为不受大气层或云层影响。

句中 it 指的是 Solar Power Satellite (太阳能发电卫星)，而 this orbit 指的是 geostationary orbit (地球静止轨道)。

Reading Material

The Characteristics of Sunlight

1. Particle-wave duality

Our understanding of the nature of light has changed back and forth over the past few centuries between two apparently conflicting viewpoints. A highly readable account of the evolution of quantum theory is given in Gribben. In the late 1600s, Newton's mechanistic view of light as being made up of small particles prevailed. By the early 1800s, experiments by both Young and Fresnel had shown **interference effects** in light beams, indicating that light was made up of waves. By the 1860s, Maxwell's theories of electromagnetic radiation were accepted, and light was understood to be part of a wide spectrum of electromagnetic waves with different wavelengths. In 1905, Einstein explained the **photoelectric effect** by proposing that light is made up of discrete particles or quanta of energy^①. This complementary nature of light is now well accepted. It is referred to as the particle-wave duality, and is summarized by the equation.

$$E = hf = hc/\lambda$$

where light, of frequency f or wavelength, comes in 'packets' or *photons*, of energy E , h is Planck's constant ($6.626 \times 10^{-34} \text{ J} \cdot \text{s}$) and c is the velocity of light ($3.00 \times 10^8 \text{ m/s}$). In defining the characteristics of photovoltaic or 'solar' cells, light is sometimes treated as waves, other times as particles or photons.

2. Black radiation

A 'blackbody' is an ideal absorber, and emitter, of radiation. As it is heated, it starts to glow; that is, to emit electromagnetic radiation. A common example is when a metal is heated. The hotter it gets, the shorter the wavelength of light emitted and an initial red glow gradually turns white. Classical physics was unable to describe the wavelength distribution of light emitted from such a heated object. However, in 1900, Max Planck derived a mathematical expression describing this distribution, although the underlying physics was not understood until Einstein's work on 'quanta' five years later.

Fig. 1.3 illustrates the radiation distribution for different blackbody temperatures, as would be observed at the surface of the blackbody. The lowermost curve is that for a body heated to 3000K, about the temperature of the tungsten filament in an incandescent lamp. The wavelength of peak energy emission is about $1\mu\text{m}$, in the infrared. Only a small amount of energy is emitted at visible wavelengths ($0.4\text{--}0.8\mu\text{m}$) in this case, which explains why these lamps are so inefficient. Much higher temperatures, beyond the melting points of most metals, are required to shift the peak emission to this range.

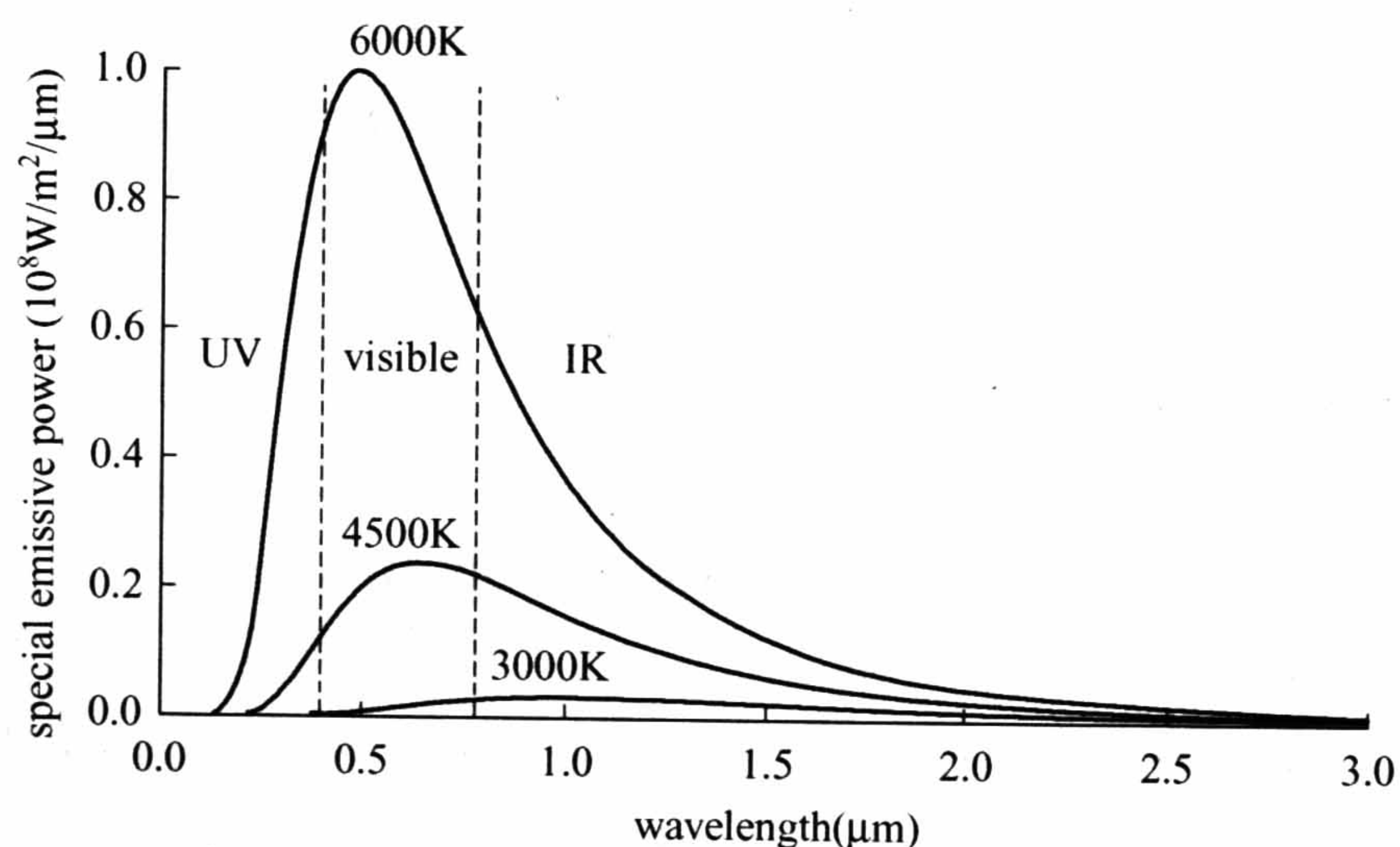


Fig. 1.3 Radiation distributions from perfect blackbodies at three different temperatures, as would be observed at the surface of the blackbodies

3. The sun and its radiation

The sun is a hot sphere of gas heated by **nuclear fusion reactions** at its centre^②. Internal temperatures reach a very warm 20 million K and the intense radiation from the interior is absorbed by a layer of hydrogen ions closer to the sun's surface. Energy is transferred by convection through this optical barrier and then re-radiated from the outer surface of the sun, the photosphere. This emits radiation approximating that from a blackbody with a temperature of nearly 6000 K, as shown in Fig. 1.4.

4. Solar radiation

Although radiation from the sun's surface is reasonably constant, by the time it reaches the earth's surface it is highly variable owing to absorption and scattering in the earth's atmosphere. When skies are clear, the maximum radiation strikes the earth's surface when the sun is directly overhead, and sunlight has the shortest **pathlength** through